Nevo and Rosen (2012): Simulations and Extensions

Maximilian Grotz
Toulouse School of Economics

April 24, 2026

1 Nevo and Rosen (2012)

2 Nevo and Rosen (2012) as Sensitivity Analysis

2.1 Sensitivity Analysis: General Approach

This material has mainly been taken from Paul Diegert's lecture slides on sensitivity analysis.

Consider a family of models indexed by a sensitivity parameter $\delta \in [\underline{\delta}, \bar{\delta}]$:

$$\mathcal{M}(\delta) \subseteq \mathcal{M}(\delta') \subseteq \mathcal{M} \quad \text{for } \delta \leq \delta'.$$

The parameter δ governs the strength of assumptions: larger δ corresponds to weaker assumptions and hence a larger model class.

Assume models are *complete*, i.e. each model $m \in \mathcal{M}$ induces a distribution F^m over observables.

Let F_0 denote the observed data distribution. The identified set under assumption δ is $\mathcal{M}_0(\delta) = \{m \in \mathcal{M}(\delta) : F^m = F_0\}$

For a parameter of interest $\psi : \mathcal{M} \to \mathbb{R}$, the identified set is

$$\Psi_0(\delta) = \{\psi(m) : m \in \mathcal{M}_0(\delta)\}.$$

As δ increases, the model class $\mathcal{M}(\delta)$ expands, so the identified set $\Psi_0(\delta)$ weakly enlarges. Thus δ can be interpreted as a *sensitivity parameter* controlling the degree of identification.

2.2 Connection to Nevo and Rosen (2012)

We can see Nevo and Rosen (2012) imposing the following restrictions with $\delta = 1$:

$$\text{sign}\Big(\text{Cov}(Z,u)\Big) = \text{sign}\Big(\text{Cov}(X,u)\Big), \qquad |\text{Corr}(Z,u)| \leq \delta |\text{Corr}(X,u)|.$$

These restrictions define a set of admissible data-generating processes consistent with observed moments and the structural parameter $\psi(m) = \beta$. This corresponds to specifying a model class $\mathcal{M}(\delta)$, where $\delta \in [0,1]$.

2.3 Identified Set in a Linear Model

2.3.1 Univariate instrument

Redoing calculations from Nevo and Rosen (2012) we find the following bounds:

$$\underline{\beta}(\delta) = \min\left\{\beta^{IV}_{V(0)}, \beta^{IV}_{V(\delta)}\right\},$$
$$\overline{\beta}(\delta) = \max\left\{\beta^{IV}_{V(0)}, \beta^{IV}_{V(\delta)}\right\}.$$

See Appendix B for the full calculation.

In Figure 1 we plot the identified set $B(\delta) = [\underline{\beta}(\delta), \overline{\beta}(\delta)]$ as a function of δ. As δ increases, the bounds widen, reflecting weaker assumptions and hence less precise identification. We can see that the true β is contained in the identified set for $\delta \geq \frac{\rho_{ZU}}{\rho_{XU}}$, as then Assumption 4 is satisfied.

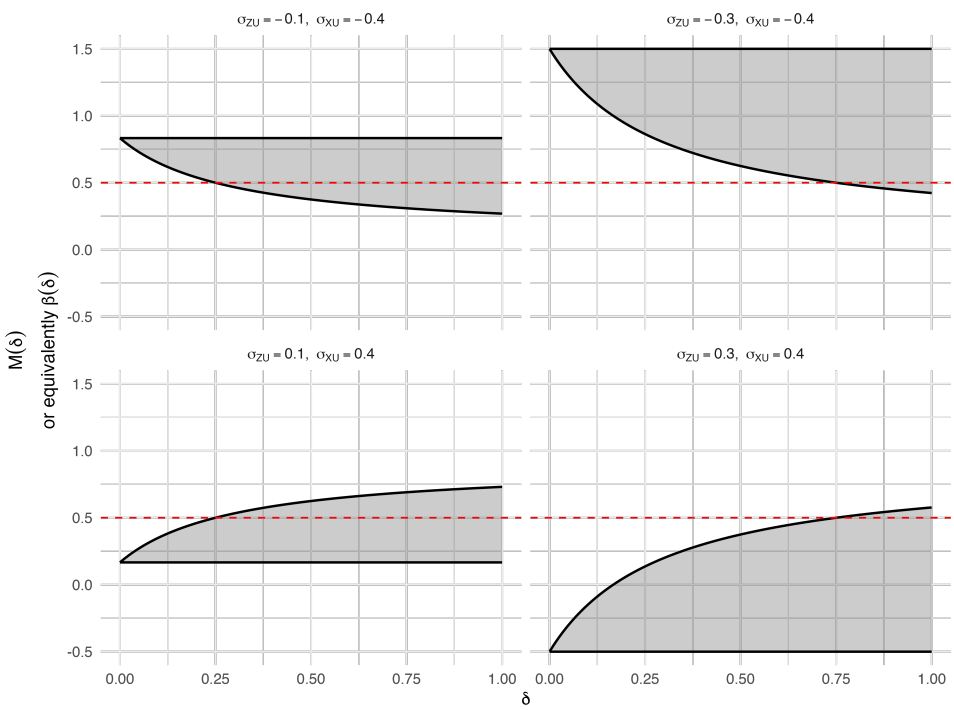

Figure 1: Identified bounds on β as a function of δ. Red line signifies true β. $\sigma_X = \sigma_Z = \sigma_U = 1$ and $\text{Cov}(X,Z) = -0.3$

2.3.2 Bivariate instrument

The identified set with two instruments is just the intersection of the identified sets from each instrument. The fomula is in B.

In Figure 2 the dark blue region corresponds to the strength of assumptions we need to make to reject $\beta = 0.4$ in our hypothetical example. The light blue area is the area we are able to reject at now that we have included a second instrument.

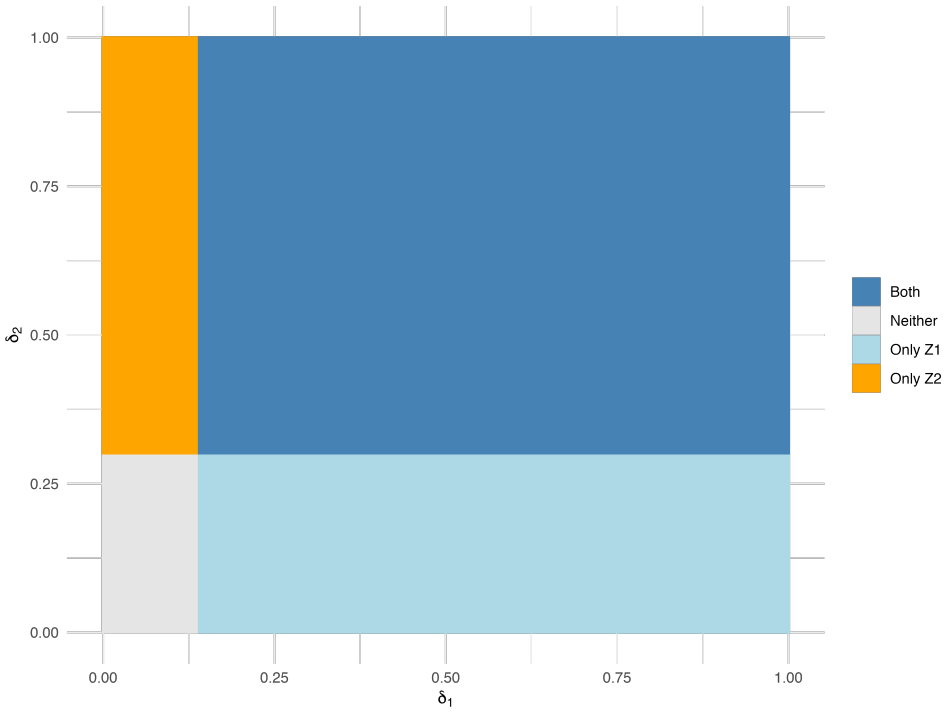

Figure 2: Parameters are set to $\mathrm{Var}(X) = \mathrm{Var}(U) = \mathrm{Var}(Z_1) = \mathrm{Var}(Z_2) = 1$, with $\beta = 0.5$, $\mathrm{Cov}(X, U) = 0.4$, $\mathrm{Cov}(Z_1, U) = 0.1$, $\mathrm{Cov}(Z_2, U) = 0.2$, $\mathrm{Cov}(X, Z_1) = -0.3$, and $\mathrm{Cov}(X, Z_2) = -0.5$. The figure shows, over a grid of (δ_1, δ_2), whether the value 0.4 lies within the identified set based on Z_1 alone, Z_2 alone, and their intersection.

3 General simulations

4 Networks

References

Nevo, A., & Rosen, A. M. (2012). Identification with imperfect instruments. *The Review of Economics and Statistics*, *94*(3), 659–671.

A An Example of Sensitivity Analysis

B Identified Set in Linear Model

B.1 Univariate Instrument

$$Y = \beta X + U, \qquad E[U] = 0.$$

Assume

$$\rho_{XU}\rho_{ZU} \geq 0, \qquad |\rho_{ZU}| \leq \delta|\rho_{XU}|, \qquad \delta \in [0,1].$$

Define

$$\lambda^* = \frac{\rho_{ZU}}{\rho_{XU}}.$$

Then

$$\lambda^* \in [0, \delta].$$

For any $\lambda \in [0, \delta]$, define

$$V(\lambda) = \sigma_X Z - \lambda \sigma_Z X.$$

The IV estimand using $V(\lambda)$ is

$$\beta^{IV}_{V(\lambda)} = \frac{\mathrm{Cov}(V(\lambda), Y)}{\mathrm{Cov}(V(\lambda), X)}.$$

Equivalently,

$$\beta^{IV}_{V(\lambda)} = \frac{\sigma_X \sigma_{ZY} - \lambda \sigma_Z \sigma_{XY}}{\sigma_X \sigma_{XZ} - \lambda \sigma_Z \sigma_X^2}.$$

Hence the identified set is

$$B(\delta) = \left\{ \beta^{IV}_{V(\lambda)} : \lambda \in [0, \delta] \right\}.$$

Since

$$\frac{d}{d\lambda} \beta^{IV}_{V(\lambda)} = \frac{\sigma_X \sigma_Z \left(\sigma_{ZY} \sigma_X^2 - \sigma_{XY} \sigma_{XZ} \right)}{\left(\sigma_X \sigma_{XZ} - \lambda \sigma_Z \sigma_X^2 \right)^2},$$

the sign of the derivative does not depend on λ. Therefore the bounds are attained at the endpoints:

$$\underline{\beta}(\delta) = \min \left\{ \beta^{IV}_{V(0)}, \beta^{IV}_{V(\delta)} \right\},$$

$$\overline{\beta}(\delta) = \max \left\{ \beta^{IV}_{V(0)}, \beta^{IV}_{V(\delta)} \right\}.$$

Because

$$V(0) = \sigma_X Z,$$

we have

$$\beta^{IV}_{V(0)} = \beta^{IV} = \frac{\sigma_{ZY}}{\sigma_{XZ}}.$$

Thus

$$\boxed{B(\delta) = \left[\min\left\{\frac{\sigma_{ZY}}{\sigma_{XZ}}, \frac{\sigma_X\sigma_{ZY} - \delta\sigma_Z\sigma_{XY}}{\sigma_X\sigma_{XZ} - \delta\sigma_Z\sigma_X^2}\right\},\ \max\left\{\frac{\sigma_{ZY}}{\sigma_{XZ}}, \frac{\sigma_X\sigma_{ZY} - \delta\sigma_Z\sigma_{XY}}{\sigma_X\sigma_{XZ} - \delta\sigma_Z\sigma_X^2}\right\}\right].}$$

B.2 Bivariate Instrument

$$B(\delta_1, \delta_2) = \left[\max\left\{\underline{B}_1(\delta_1), \underline{B}_2(\delta_2)\right\},\ \min\left\{\overline{B}_1(\delta_1), \overline{B}_2(\delta_2)\right\}\right],$$

with

$$\underline{B}_j(\delta_j) = \min\left\{\frac{\sigma_{Z_jY}}{\sigma_{XZ_j}}, \frac{\sigma_X\sigma_{Z_jY} - \delta_j\sigma_{Z_j}\sigma_{XY}}{\sigma_X\sigma_{XZ_j} - \delta_j\sigma_{Z_j}\sigma_X^2}\right\},$$

and

$$\overline{B}_j(\delta_j) = \max\left\{\frac{\sigma_{Z_jY}}{\sigma_{XZ_j}}, \frac{\sigma_X\sigma_{Z_jY} - \delta_j\sigma_{Z_j}\sigma_{XY}}{\sigma_X\sigma_{XZ_j} - \delta_j\sigma_{Z_j}\sigma_X^2}\right\}.$$